Which language for which use in quant finance?

Python, VBA, C++, C#, R, Matlab: each one has its own natural playing field.

COMPILED VS INTERPRETED

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Compiled (C++/C#): faster at runtime  |  Interpreted (Python/R): faster to write and test
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PYTHON

- The dominant language in quant research: pandas, NumPy, scikit-learn.
- Fast to write, with a huge ecosystem of finance and data libraries.
- Slower at pure execution than C++, but good enough for most use cases.

VBA

- Automates Excel, still very common in front-office and reporting.
- Easy to pick up, but poorly suited to large data volumes or heavy computation.
- Often the first programming language learned in finance.

C++

- The reference for high-performance pricing and low-latency trading.
- Fine-grained memory control, very fast execution.
- Longer and more demanding to develop than Python.

R & MATLAB

- Historically strong in statistics and academic/research settings.
- R excels at econometrics, Matlab at matrix computation and rapid prototyping.
- Increasingly challenged by Python in industry, still present in research.

Who uses what?

QUANT RESEARCHER
Python to prototype signals, backtest and analyze data.

FRONT-OFFICE ANALYST
VBA to automate Excel models for reporting or simple valuation.

PRICING DEVELOPER
C++ or C# to build performant calculation engines in production.

RESEARCHER / ACADEMIC
R or Matlab for statistical and econometric modeling.

KEY TAKEAWAYS

1. Python has become the default language for quant research, though not necessarily for low-latency production systems.
2. VBA is underrated: still massively used in practice despite its technical limits.
3. The choice of language mostly depends on which constraint dominates: development speed (Python/VBA) vs execution speed (C++/C#).

Comparison table

LANGUAGE	EXECUTION SPEED	LEARNING CURVE	TYPICAL USE CASE
Python	Moderate	Low	Quant research, data, ML
VBA	Low	Very low	Excel automation
C++	Very high	High	High-performance pricing, trading
C#	High	Moderate	Front-office/desktop applications
R / Matlab	Moderate	Moderate	Statistics, econometrics, research

Real-world examples

PANDAS FOR BACKTESTING

A quant researcher uses pandas to manipulate price series and test a strategy on 10 years of data in a few lines of code.

VBA REPORTING MACRO

A macro automates the daily update of a risk report directly in a shared Excel file.

C++ PRICING ENGINE

An options pricing engine written in C++ computes thousands of prices per second to feed a real-time trading system.

ECONOMETRIC MODEL IN R

A researcher estimates a GARCH model in R to publish a study on emerging market volatility.

Quick glossary

VECTORIZATION
Applying an operation to an entire array rather than element by element, faster in Python.

COMPILED VS INTERPRETED
A compiled language is translated to machine code before running; an interpreted one runs on the fly.

LIBRARY
Set of reusable functions (e.g. pandas, NumPy) that saves you from reinventing the wheel.

IDE
Integrated development environment, makes writing and debugging code easier.

NOTEBOOK
Interactive document mixing code, results and text, widely used in quant research (Jupyter).

API
Interface that lets a program interact with another piece of software or service.

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